LESSON 01 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Harmonic progressions are **not** in the official JEE Advanced syllabus (which lists AP, GP, AM/GM and the power sums only). They appear in IOQM/RMO-style material and older JEE Main contexts. The technique — reciprocate, solve the AP, reciprocate back — is the transferable part.

HP via reciprocals

KEY RESULT

Definitions and the only method

A **harmonic progression** is a sequence whose reciprocals form an AP. There is no closed formula for its n-th term or sum comparable to AP/GP: every HP problem is routed through the reciprocal AP.

The **harmonic mean** of $a, b > 0$ is $H = \frac{2ab}{a+b} = \frac{2}{\frac{1}{a}+\frac{1}{b}}$ — defined by $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ in AP. For n numbers, $H = \frac{n}{\sum 1/a_i}$.

WORKED EXAMPLE

Worked — HP basics

HM of 2 and 4: $H = \frac{2\cdot 2\cdot 4}{6} = \frac{8}{3}$. The HP $1, \frac{1}{2}, \frac{1}{3}, \dots$ has reciprocals $1, 2, 3, \dots$ (an AP), so its 4-th term is $\frac{1}{4}$. General HP term: if the reciprocal AP has first term $\frac{1}{a}$ and difference d', the n-th HP term is $\frac{1}{\frac{1}{a}+(n-1)d'}$.

KEY RESULT

The chain A ≥ G ≥ H and the identity G² = AH

For two positive numbers, $A = \frac{a+b}{2}$, $G = \sqrt{ab}$, $H = \frac{2ab}{a+b}$ satisfy $A \geq G \geq H$, and moreover

$$G^2 = AH$$

since $\frac{a+b}{2} \cdot \frac{2ab}{a+b} = ab$. So A, G, H are in GP. Equality throughout iff $a = b$. For n positives, $A \geq G \geq H$ still holds (apply AM–GM to the numbers and to their reciprocals).

STANDARD APPLICATION

Standard applications

Average speed over equal distances at speeds u, v is the **harmonic** mean $\frac{2uv}{u+v}$, not the arithmetic mean — the canonical HP-in-disguise. Work-rate and parallel-resistance formulas are the same shape: whenever the reciprocals add, H appears.

COMMON TRAP

Common traps — HP

- Inventing an "S_n formula" for HP by analogy with AP/GP: none exists in elementary closed form; sums route through $H_n = \sum \frac{1}{k}$ estimates only.
- H for numbers of mixed sign: the mean framework assumes positives.
- Confusing "insert harmonic means" (reciprocate, insert AMs, reciprocate back) with inserting AMs directly.

LESSON 02 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Linear recurrences and generating functions sit outside the official AP/GP syllabus lines; they are the standard Olympiad machinery for sequences defined inductively.

First- and second-order recurrences

SHORT DERIVATION

Fixed-point shift for $a_n = p\cdot a_{n-1} + q$

Seek L with $L = pL + q$, i.e. $L = \frac{q}{1-p}$ $(p \neq 1)$. Then $a_n - L = p(a_{n-1} - L)$: a GP in disguise, so $a_n = L + (a_1 - L)p^{n-1}$. Worked: $a_n = 2a_{n-1} + 1$, $a_1 = 1$: $L = -1$, $a_n = 2^n - 1$ ✓ $(1, 3, 7, 15, \ldots)$. If $p = 1$ it is an AP instead.

KEY RESULT

Characteristic equation for $a_n = p\cdot a_{n-1} + q\cdot a_{n-2}$

Try $a_n = r^n$: the **characteristic equation** is $r^2 - pr - q = 0$. Distinct roots $r_1 \neq r_2$: $a_n = Ar_1^n + Br_2^n$; repeated root r: $a_n = (A + Bn)r^n$. Constants from two initial values. Always verify the closed form against the first computed terms.

Generating functions

KEY RESULT

The clothesline: operations on $G(x)$ are operations on the sequence

$G(x) = \sum_{n \geq 0} a_n x^n$. Multiply by x: shift right. Multiply by $\frac{1}{1-x}$: replace a_n by its partial sums. Differentiate: multiply a_n by n. $G(-x)$: alternate signs. Recurrences become algebraic equations for $G(x)$; solving and expanding reads off a_n.

SHORT DERIVATION

Fibonacci: $G(x)$ to Binet in three moves

With $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$: comparing coefficients of x^n $(n \geq 2)$ in $F(x)$ and $xF(x) + x^2F(x) + x$ gives

$$F(x) = \frac{x}{1 - x - x^2} = \frac{x}{(1 - \varphi x)(1 - \psi x)}, \qquad \varphi, \psi = \frac{1 \pm \sqrt{5}}{2}$$

Partial fractions: $F(x) = \frac{1}{\sqrt{5}}\left(\frac{1}{1 - \varphi x} - \frac{1}{1 - \psi x}\right)$; expanding both geometric series,

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$$

Checks: $F_1 = \frac{2.236068}{2.236068} = 1$ ✓; $F_7 = 13$ ✓; $F_{10} = 55$ ✓. This is Binet with no induction — pure algebra (and it matches the quadratic route of chapter 6, Olympiad sheet 01).

COMMON TRAP

Common traps — recurrences and generating functions

- Characteristic equation sign: for $a_n = pa_{n-1} + qa_{n-2}$ it is $r^2 - pr - q = 0$, not $r^2 + pr + q$.
- Repeated-root case forgotten: $(A + Bn)r^n$, not Ar^n.
- Closed form not checked at $n = 1, 2$: index-shift errors are invisible otherwise.
- Generating-function manipulations treated as convergent series: for coefficient extraction they are formal power series; convergence is a separate question.

LESSON 03 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Multi-factor telescopes, infinite products, discrete l'Hôpital and growth-rate answers are Olympiad/analysis-flavoured extensions of the Part I summation engines.

Products telescope too

SHORT DERIVATION

The product $\prod(1 - 1/k^2)$

Since $\log\prod a_k = \sum \log a_k$, products collapse by the same cancellation as sums. The unlocking identity is the difference of squares: $1 - \frac{1}{k^2} = \frac{k-1}{k}\cdot\frac{k+1}{k}$. Hence

$$\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right) = \prod_{k=2}^{n}\frac{k-1}{k}\cdot\prod_{k=2}^{n}\frac{k+1}{k} = \frac{1}{n}\cdot\frac{n+1}{2} = \frac{n+1}{2n} \to \frac{1}{2}$$

Check at $n=4$: $\frac{3}{4}\cdot\frac{8}{9}\cdot\frac{15}{16} = \frac{5}{8} = \frac{4+1}{2\cdot 4}$ ✓. The same split in $\prod\frac{4k^2}{4k^2-1} = \prod\frac{2k}{2k-1}\cdot\frac{2k}{2k+1}$ gives the finite closed form $\frac{2^{4n}(n!)^4}{(2n+1)((2n)!)^2}$, whose limit is **Wallis's product** *$\frac{\pi}{2}$.*

Stolz–Cesàro: l'Hôpital for sequences

KEY RESULT

The discrete l'Hôpital

If b_n is strictly increasing and unbounded, then $\lim\frac{a_n}{b_n} = \lim\frac{a_{n+1}-a_n}{b_{n+1}-b_n}$ whenever the latter limit exists. No differentiability needed.

Worked: $\lim\frac{H_n}{\ln n}$ with $H_n = \sum_1^n \frac{1}{k}$: the difference ratio is $\frac{1/(n+1)}{\ln(1+1/n)} \to 1$, so $\frac{H_n}{\ln n} \to 1$.

Numerically $1.1264, 1.0836, 1.0627, 1.0501$ at $n = 10^2, \ldots, 10^5$ — approaching 1 slowly, at rate $\sim \frac{\gamma}{\ln n}$ with γ Euler's constant.

Asymptotics: when the honest answer is a growth rate

WORKED EXAMPLE

Worked — $a_{n+1} = a_n + 1/a_n$ grows like $\sqrt{(2n)}$

$a_1 = 1$, $a_{n+1} = a_n + \frac{1}{a_n}$. Square: $a_{n+1}^2 = a_n^2 + 2 + \frac{1}{a_n^2}$, telescoping to $a_n^2 = 2(n-1) + \sum_{k=1}^{n-1} \frac{1}{a_k^2} + 1$. Since $a_k^2 \approx 2k$, the terms $\frac{1}{a_k^2} \approx \frac{1}{2k}$, so the sum grows only like $\frac{1}{2}\ln n$; hence $a_n^2 = 2n + O(\log n)$ and $a_n \sim \sqrt{2n}$. (Note: the remainder sum does **not** converge to a constant — it diverges logarithmically; the growth-rate conclusion is unaffected.) Numerics: $a_{10} = 4.5699$ vs $\sqrt{20} = 4.4721$, ratio $\to 1$ ✓. The method: **transform until it telescopes, then estimate the remainder.**

COMMON TRAP

Common traps — advanced summation

- Infinite product "cancellation" without checking convergence to a **nonzero** limit.
- Stolz–Cesàro applied with non-monotone denominator, or in the converse direction (the difference-limit may fail while the ratio-limit exists).
- Asymptotic claims stated as equalities: $a_n \sim \sqrt{2n}$ means ratio $\to 1$, not $a_n = \sqrt{2n} + o(1)$.
- Telescoping arctan-type sums (e.g. $\sum \arctan \frac{1}{k^2+k+1} = \frac{\pi}{4} - \arctan \frac{1}{n+1}$) need branch care: each difference must land in the principal range (chapter 4 of this series).